Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

Course Title: Basic Electrical and Electronic Circuit

Course Code: EEE 0713 1206B

Experiment 06

Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit

---

Submitted By:
Raj Bhowmik
Reg. No: 202561901048
Session: 2024-25
Department of Software Engineering

Submitted To:
Md Rasedujjaman
Associate Professor
Department of Electrical and Electronic Engineering
Shahjalal University of Science and Technology

Submission Date: August 25, 2026

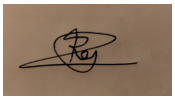
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: August 25, 2026



Raj Bhowmik
Reg. No: 202561901048

Contents

Declaration i

List of Figures iii

List of Tables iv

Objective 1

Introduction 1

Theory 1
- Conduction Analysis 1
- Mathematical Formulation 1
- Parallel Capacitive Filter 2

Apparatus and Components 2

Circuit Diagram 2

Experimental Procedure 2

Theoretical Calculation 3

Data Collection 3

Result Analysis 4

Discussion and Sources of Error 4

Conclusion 4

References 4

List of Figures

1 Full-wave bridge rectifier circuit incorporating transformer feed, four-diode bridge array, load resistor R_L, and optional parallel filter capacitor C. 3

List of Tables

1 List of Equipment and Components Used 2
2 Unfiltered Rectifier Performance Across Load Settings 3
3 Filtered Rectifier Performance: Smoothing Effect of C 4

Objective

- To design, construct, and evaluate a full-wave bridge rectifier circuit on a breadboard using silicon diodes.
- To examine the cycle-by-cycle conduction path during positive and negative input AC half-cycles.
- To record input and output voltage waveforms using a Digital Storage Oscilloscope (DSO).
- To evaluate key performance parameters, including average DC output voltage (V_{DC}), RMS output voltage (V_{rms}), ripple factor (r), and rectification efficiency (η).
- To assess output smoothing achieved via parallel capacitive filters under varying load conditions.

Introduction

Rectification convert bidirectional Alternating Current (AC) into unidirectional Direct Current (DC). Among standard topologies, full-wave bridge rectifiers offer superior utilization of AC supply cycles without requiring a center-tapped transformer. Arranging four diodes in a bridge array directs current through the load resistor in a single physical direction across both input half-cycles. The output waveform is subsequently smoothed using parallel capacitive filtering to deliver stable DC levels appropriate for electronic circuitry.

Theory

Conduction Analysis

Consider the AC voltage delivered by the step-down transformer:

$$v_s(t) = V_m \sin(\omega t),$$

where $V_m = V_{rms}\sqrt{2}$ is peak secondary voltage and $\omega = 2\pi f$ is the supply angular frequency.

- **Positive Half-Cycle ($0 < \omega t < \pi$):** The upper secondary terminal operates at higher potential relative to the lower terminal. Diodes D_1 and D_4 are forward-biased, while D_2 and D_3 remain reverse-biased. Current travels through D_1, downward through load R_L, and returns via D_4.
- **Negative Half-Cycle ($\pi < \omega t < 2\pi$):** Supply polarity reverses. Diodes D_2 and D_3 transition to forward conduction, whereas D_1 and D_4 become reverse-biased. Current passes through R_L in the identical physical direction (top-to-bottom) via D_2 and D_3.

Because current flows through the load in both half-cycles, the output ripple frequency is double the input frequency ($f_{\text{out}} = 2f_{\text{in}}$).

Mathematical Formulation

Accounting for the forward voltage drop ($V_D \approx 0.7\,\text{V}$) across two conducting diodes in each half-cycle, peak load voltage is:

$$V_{mo} = V_m - 2V_D.$$

The average DC output voltage and RMS output voltage are calculated as:

$$V_{DC} = \frac{2V_{mo}}{\pi}, \qquad V_{rms} = \frac{V_{mo}}{\sqrt{2}}.$$

The ripple factor (r), representing relative AC content in the rectified output, is expressed by:

$$r = \sqrt{\left(\frac{V_{rms}}{V_{DC}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} \approx 0.482.$$

Theoretical maximum rectification efficiency and peak inverse voltage (PIV) rating per diode are:

$$\eta_{\max} = \frac{8}{\pi^2} \approx 81.2\%, \qquad \text{PIV} = V_m.$$

Parallel Capacitive Filter

Connecting a filter capacitor C in parallel with load R_L smooths the output profile by charging rapidly near voltage peaks and discharging through the load between cycles. Peak-to-peak ripple voltage ($V_{r(p\text{-}p)}$) and overall ripple factor (r) are modeled as:

$$V_{r(p\text{-}p)} \approx \frac{I_{DC}}{2fC} = \frac{V_{DC}}{2fR_LC}, \qquad r \approx \frac{1}{4\sqrt{3}\,fR_LC}.$$

Increasing either capacitance C or load resistance R_L minimizes output voltage ripple.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
Step-Down Transformer	230 V/12 V RMS, 50 Hz	1
Silicon Rectifier Diodes	1N4007	4
Load Resistor R_L	1 kΩ, 2.2 kΩ	2
Electrolytic Filter Capacitor C	47 µF, 100 µF	2
Digital Multimeter (DMM)	DC/AC Volts	1
Digital Storage Oscilloscope (DSO)	Dual-Channel with 10× Probes	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

Circuit Diagram

Experimental Procedure

1. Construct the bridge configuration using four 1N4007 diodes on the breadboard, observing correct terminal polarity.
2. Connect the 12 V secondary terminals of the transformer to the AC input terminals (AC_1, AC_2) of the bridge network.
3. Connect $R_L = 1\,\text{k}\Omega$ across the DC output without capacitive filtering. Observe AC input and rectified DC output signals on Channel 1 and Channel 2 of the DSO respectively. Measure V_{DC} and V_{rms} using the DMM.
4. Connect filter capacitor $C = 47\,\mu\text{F}$ across R_L. Record output V_{DC} and peak-to-peak ripple voltage $V_{r(p\text{-}p)}$ on the DSO.
5. Repeat filtered measurements for $C = 100\,\mu\text{F}$ and load setting $R_L = 2.2\,\text{k}\Omega$ to evaluate ripple suppression behavior.

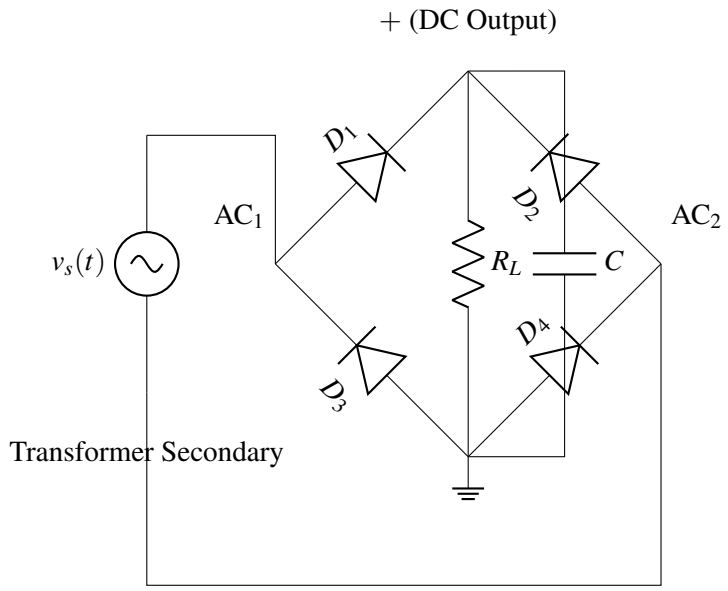


Figure 1: Full-wave bridge rectifier circuit incorporating transformer feed, four-diode bridge array, load resistor R_L, and optional parallel filter capacitor C.

Theoretical Calculation

Using supply voltage $V_{rms} = 12\,\text{V}$, frequency $f = 50\,\text{Hz}$, diode drop $V_D = 0.7\,\text{V}$, and load $R_L = 1\,\text{k}\Omega$:

$$V_m = V_{rms}\sqrt{2} = 12 \times 1.4142 = 16.97\,\text{V},$$

$$V_{mo} = V_m - 2V_D = 16.97 - 1.4 = 15.57\,\text{V},$$

$$V_{DC} = \frac{2V_{mo}}{\pi} = \frac{2 \times 15.57}{3.1416} = 9.91\,\text{V}, \qquad I_{DC} = \frac{V_{DC}}{R_L} = \frac{9.91}{1000} = 9.91\,\text{mA},$$

$$V_{rms(\text{out})} = \frac{V_{mo}}{\sqrt{2}} = \frac{15.57}{1.4142} = 11.01\,\text{V},$$

$$\eta = \frac{V_{DC}^2/R_L}{V_{rms(\text{out})}^2/R_L} \times 100\% = \frac{9.91^2}{11.01^2} \times 100\% \approx 81.0\%, \qquad \text{PIV} = V_m = 16.97\,\text{V}.$$

With filter capacitor $C = 47\,\mu\text{F}$ connected:

$$V_{r(p\text{-}p)} \approx \frac{V_{DC}}{2fR_LC} = \frac{9.91}{2 \times 50 \times 1000 \times 47 \times 10^{-6}} \approx 2.11\,\text{V}.$$

Data Collection

Table 2: Unfiltered Rectifier Performance Across Load Settings

Load R_L	Peak Input V_m	Peak Output V_{mo}	Measured V_{DC}	Measured V_{rms}	Ripple Factor r	Efficiency η
1 kΩ	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
2.2 kΩ	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

Note on Experimental Data

Laboratory measurements will be populated upon final validation of the experimental data sheet. All formatting parameters and analytical frameworks are fully prepared to evaluate collected parameters.

Table 3: Filtered Rectifier Performance: Smoothing Effect of C

Load R_L	**Filter C**	**Measured V_{DC}**	**Ripple $V_{r(p\text{-}p)}$**
1 kΩ	47 µF	*TBD*	*TBD*
1 kΩ	100 µF	*TBD*	*TBD*
2.2 kΩ	47 µF	*TBD*	*TBD*

Result Analysis

Experimental readings will be compared against theoretical formulations using percentage error:

$$\% \text{ Error} = \left| \frac{\text{Theoretical} - \text{Measured}}{\text{Theoretical}} \right| \times 100\%.$$

Output ripple frequency will be evaluated against $f_{\text{out}} = 2f_{\text{in}} = 100\,\text{Hz}$. The reduction in peak-to-peak ripple voltage resulting from higher capacitance values ($C = 100\,\mu\text{F}$) will be verified against theoretical models.

Discussion and Sources of Error

Deviations between experimental and theoretical parameters stem from the following technical factors:

1. **Diode Forward Drop Variations:** Actual forward voltage drop across 1N4007 diodes under load may vary from the nominal 0.7 V model.
2. **Transformer Regulation Loss:** Internal transformer winding resistance causes output voltage droop under load conditions.
3. **Capacitor Tolerances and ESR:** Electrolytic capacitor tolerance ranges and non-zero Equivalent Series Resistance (ESR) alter peak ripple attenuation.
4. **Measurement Equipment Loading:** DSO probe input impedance and minor AC line frequency variations introduce small experimental errors.

Conclusion

The design and operational mechanics of the full-wave bridge rectifier were systematically evaluated. The circuit effectively doubles ripple frequency relative to input AC supply and increases DC conversion efficiency. The integration of parallel capacitive filtering demonstrates substantial attenuation of AC ripple components, delivering a stabilized DC voltage profile.

References

1. Boylestad, R. L., & Nashelsky, L. (2013). *Electronic Devices and Circuit Theory* (11th ed.). Pearson.
2. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
3. Sedra, A. S., & Smith, K. C. (2014). *Microelectronic Circuits* (7th ed.). Oxford University Press.